

Chapter 56 — Hypospadias

- **Duckett (1995):** “Hypospadiology” demands study of predecessors and current experts, and **training in high-volume centres beats learning from one's own mistakes.**
- Progress continues to depend on **meticulous documentation, multicentre collaboration and honest outcome reporting.**

EMBRYOLOGY

- **Classic triad:** **dorsally hooded foreskin, proximal (ventral) meatus, ventral penile curvature**, all from **arrested penile development.**
- **Indifferent stage:** mesenchyme divides the cloacal membrane; cloacal folds merge into the **genital tubercle**. Anal and genital systems separate at **7 weeks**. Urogenital folds and, laterally, labioscrotal folds flank the urogenital membrane.
- **Critical window: weeks 8–16**, with urethral closure proceeding **from the anus towards the distal glans.**

	Ectodermal ingrowth theory (Glenister, 1954)	Endodermal theory (Kurzrock): “two-zipper”
Glanular urethra	Arises separately, by retrograde ectodermal ingrowth from the glans tip, fusing at the corona with the endodermal penile urethra	Entirely endodermal: the plate is continuous through the glans, which differentiates into stratified squamous epithelium
Mechanism	Fusion of urethral folds	“Opening zipper” (plate canalises into a groove; both sexes) then “closing zipper” (androgen-dependent, male only) , with folds interlacing at varying rates and levels
Status	Classic	Current evidence favours this; its disruption explains the phenotypic spectrum

- **Prepuce:** folds appear at about **8 weeks**, fuse **dorsally first**, then ventrally. **Ventral fusion awaits urethral fold fusion (about 20 weeks)**, so an ectopic meatus comes with an **incomplete foreskin in 95%.**
- **Megameatus with intact prepuce (MIP) (5%):** a wide distal meatus, normal foreskin, no severe curvature. The preputial and urethral processes can therefore dissociate.
- **Hormonal phases:** a hormone-independent phase, then from **week 8** a hormone-dependent one. **SRY** (sex-determining region Y) at **7–8 weeks** → testis → testosterone → regression of female structures and tubercle elongation.
- Leydig-cell testosterone → **5 α -reductase type 2** → **dihydrotestosterone (DHT)** locally in the genitalia. **Sonic hedgehog (Shh)** drives tubercle and plate formation.
- **Androgen receptors** lie in penile skin, inner prepuce, urethra and spongiosal stroma. Without receptor stimulation, scrotal and urethral folds fail to fuse, as in female development.

KEY POINTS

- **DHT at 8–12 weeks** is the key mediator of penile development.

- **Male urethral closure is androgen-dependent**, unlike female urethral formation.

Concept — Hypospadias is **an arrest of the androgen-dependent closing zipper**. The meatus stops where closure stopped, the prepuce cannot close ventrally over an open urethra, and the unfused ventral tissue tethers the shaft into curvature.

DIAGNOSIS OF HYPOSPADIAS

- **One of the commonest congenital anomalies: 1 in 150–300 male births.** Diagnosis is clinical: ectopic ventral meatus, ventral curvature, dorsally hooded foreskin. The last two are not always present.
- **MIP (up to 5%)** is hidden by an intact foreskin (Fig. 56.3). **Retract the foreskin before every newborn circumcision.** Previous circumcision did not harm later MIP repair, but **avoid circumcision if there is curvature or ventral skin deficiency**, because the foreskin may be needed.
- **Curvature without hypospadias** (orthotopic meatus): **defer circumcision until the curvature is repaired.**
- **Severity by meatal position: distal 70–85%, proximal 10–25%.** This **oversimplifies**: an apparently distal meatus may overlie **hypoplastic tissue and absent spongiosum to the penoscrotal junction**, with severe curvature (Fig. 56.5). **Classify after degloving**, by meatal position plus curvature.

FIG. 56.6 THE GLANS–MEATUS–SHAFT (GMS) SCORE

Score	Glans (G)	Meatus (M)	Shaft (S): curvature
1	Good size; healthy, deeply grooved plate	Glanular	None
2	Adequate; grooved plate	Coronal sulcus	Mild (<30°)
3	Small; narrow, fibrotic or flat plate	Mid or distal shaft	Moderate (30–60°)
4	Very small; indistinct, very narrow or flat plate	Proximal shaft or penoscrotal	Severe (>60°)

From Merriman LS, J Pediatr Urol 2013.

- **What it does:** scores the three components 1–4 and adds them (total 3–12). **Higher GMS scores predict more complications.** It is scored **in the operating room**, because office assessment, especially of curvature, is less accurate. Machine learning has not yet replaced the surgeon's examination.
- **The trap:** the meatus is only one of three components. A coronal meatus with a flat, small glans and 45° curvature scores worse than its position suggests.
- **History and examination: meatal stenosis**, even in distal cases, can justify surgery (“laser-beam” or pencil-thin deflected stream, Fig. 56.7). Associated **torsion, penoscrotal webbing or transposition** add complexity, as does dysplastic ventral tissue.
- **Proximal cases** have **more curvature, a shorter penis and less elastic tissue on elastography**. Ultrasound can show the **divergence of the spongiosum**.
- **Associations: inguinal hernia or hydrocele 9–16%**, not related to severity. **Cryptorchidism about 7%, nearly 10% in proximal hypospadias.**

Disorder of sex development (DSD): when to investigate	Work-up
AUA cryptorchidism guideline: unilateral or bilateral undescended testis plus severe proximal hypospadias ; also perineal hypospadias or ambiguous genitalia . Since 2006 , severe hypospadias counts as 46,XY DSD . Risk is highest with a non-palpable testis (DSD about 50% vs 15%) and with proximal vs distal hypospadias (64% vs 7%)	Ultrasound (Müllerian remnants), karyotype , testosterone , follicle-stimulating hormone (FSH) , luteinising hormone (LH) , anti-Müllerian hormone (AMH) . A 46,XY karyotype can mislead: consider 5α-reductase deficiency , androgen insensitivity , testicular dysgenesis , ovotesticular DSD

KEY POINTS

- Diagnosis is clinical: **ventral meatus, ventral curvature, dorsal hood.**
- Severity is best judged **intraoperatively, by meatus plus curvature after degloving.**
- **Hypospadias plus cryptorchidism → investigate for DSD.**

Concept — The meatus is the **least reliable** marker of severity. The true phenotype (plate, spongiosum, curvature) appears only after degloving, and the hypospadias–cryptorchidism pair is the **signal of a systemic androgen problem, not a local one.**

ETIOLOGY OF HYPOSPADIAS

- **Multifactorial.** Risk factors: **prematurity, small for gestational age** (<10th centile), **intrauterine growth restriction.** Prematurity (<37 weeks) and low birth weight cluster with **proximal hypospadias plus cryptorchidism.**
- **Placental insufficiency:** low placental weight → about **twofold** hypospadias and 1.5-fold cryptorchidism. **Assisted reproduction is an independent risk factor,** even after adjusting for prematurity, low birth weight and twinning.
- **Familial:** twin concordance **9–27%;** first-degree relative affected in **9–15%.** A Danish national registry showed **maternal and paternal transmission** and risk even in third-degree relatives.
- **No single gene:** chromosomal abnormalities in **up to 30%.** Gene categories:
 - **penile development:** HOX, fibroblast growth factor (FGF)
 - **gonadal development:** WT1, SOX9, DMRT1, GATA4
 - **androgen cascade:** LH receptor, **5 α -reductase, androgen receptor,** testicular development
- **Nearly 200 syndromes** include hypospadias, notably **WAGR** (Wilms tumour, aniridia, genitourinary anomalies, developmental delay), **Denys–Drash** (genital malformation, renal failure, high Wilms risk) and **Smith–Lemli–Opitz** (heart, lung, kidney, gut and genital malformations).
- **Incidence:** reports of rises (up to 3% per year since the 1970s) are disputed, possibly reflecting **inclusion of milder variants.** A 40-year Swedish series shows a true rise in distal and proximal cases. Overall the rate is **stable or increasing.**
- **Testicular dysgenesis syndrome (TDS):** **poor semen quality, testicular cancer, cryptorchidism and hypospadias** share a **fetal exposure disrupting the androgen cascade,** with genetic and environmental causes.
- **Endocrine-disrupting chemicals (EDCs):** pesticides, hormones (contraceptives, diethylstilbestrol (DES)), **phthalates** (plastics, hairspray), phytoestrogens (vegetarian diet). They **competitively inhibit the androgen receptor.** The link is proven in animals but unproven in humans. One case-control study found risk from **parental exposure to detergents, pesticides and cosmetics,** highest when combined with occupational exposure.
- **Anogenital distance (AGD):** centre of anus to the caudal penoscrotal junction. It is a marker of disrupted androgen action: **shorter with EDC exposure,** and linked to infertility, cryptorchidism and hypospadias. **Shorter AGD correlates with greater severity.**

KEY POINTS

- **Assisted reproduction and monozygotic twins** raise risk.
- Causes include **placental insufficiency, androgen disruption** (partly environmental) and **genetics**. **AGD reflects androgen disruption and tracks severity.**

Concept — Every aetiological thread (placenta, genes, chemicals, AGD) converges on **insufficient androgen action during weeks 8–16**. That is why hypospadias travels with cryptorchidism and poor spermatogenesis, and why the most severe cases need a DSD work-up.

SURGICAL ASSESSMENT

- **Goals: a straight erection of adequate length; a urethra to the tip** for urine and semen; **an acceptable cosmetic result**. Achieve them with the **fewest procedures** and a durable result into adulthood.
- **A straight penis with a meatus distal enough to void standing may gain little from repair**. Counsel honestly.
- **Timing: genital awareness begins at about 18 months**. Repair **before 12 months** gives less anxiety, better psychosexual outcome and possibly fewer complications.
- **AAP Section on Urology (1996): elective repair at 6–12 months**. Adult repair was once thought riskier, but recent series show **similar success**.
- **Anaesthesia**: animal neurotoxicity data exist, but children with about **1 hour of general anaesthesia** (hernia repair) showed **no neurodevelopmental delay at 2 years**. A single short exposure appears low-risk. Discuss **deferring until the child can assent** in mild cases. Unrepaired distal cases report **urinary bother and concern about appearance**.
- **History**: previous surgery, urinary tract infection (UTI), voiding dysfunction, curvature with erection, asthma, bleeding disorders. **Examination: stretched penile length** (exclude micropenis), **both groins** (undescended testis), AGD (androgen insensitivity).
- **No routine voiding cystourethrogram (VCUG) or renal ultrasound**. Associated VUR, hydronephrosis and renal ectopia are of little significance. Screen with ultrasound only if there are **other systemic anomalies**.
- **Prostatic utricle**: commoner in proximal cases. Large utricles risk **stasis, UTI and difficult catheterisation**. Endoscopic or robotic removal is possible, but most are asymptomatic and the **vas is at risk**, so intervention is rare.
- **Plan for staging** before the day of surgery. If found at newborn circumcision, **abort the circumcision** if there is curvature or skin deficiency.

Preoperative Androgen Stimulation

- **Glans width <14 mm is associated with more complications**, though this is disputed (technical error may matter more). **Dihydrotestosterone, human chorionic gonadotrophin (hCG) or testosterone** enlarges the glans and penis, reducing glansplasty tension and giving more tissue.
- **Side effects**: aggression, more erections, skin darkening, transient secondary sexual features, all resolving **within 6 months**. Theoretical concerns: **more bleeding** (angiogenesis) and **impaired healing**. **Severe proximal cases may be partly androgen-insensitive** and respond poorly.
- **Evidence**: meta-analyses are inconclusive because the data are heterogeneous. **Two randomised trials showed fewer complications** with preoperative testosterone, and surveys show common use for **proximal hypospadias or a small glans**.

- **Authors' regimen:** testosterone cypionate 2 mg/kg intramuscularly at 5 and 2 weeks before surgery if the glans width is <15 mm. This gives about 4 mm glans gain with two doses (2 mm with one), with fewer complications in formal urethroplasty.

Intraoperative Assessment and Management

REGIONAL ANAESTHESIA OPTIONS

Block	Advantages	Drawbacks
Caudal (most popular)	Less opioid; better early pain control; can be re-dosed at the end of long cases	Leg weakness (less suitable in older children). Fistula association is disputed: one study found more fistulae, attributed to penile engorgement (volume up about a quarter); others found none after controlling for surgeon and repair
Penile (circumferential)	Better pain control after distal repair in one study	Narrower coverage
Pudendal	Wider coverage (ventral penis, perineum, scrotum); no leg weakness, so suits older children	Emerging technique

- **Ketorolac 0.5 mg/kg IV** once at the end, with minimal bleeding risk.
- **Epinephrine** (topical or injected) reduces blood loss; the authors prefer a **tourniquet**.
- Give **IV antibiotics** at induction. **Cystoscopy only for severe proximal or redo cases:** a utricle can block the catheter, and anatomical variants can be found.
- Under anaesthesia: free preputial adhesions; assess the meatus, ventral tissue, glans groove depth and curvature. **A proximal hypospadias can masquerade as distal.**
- **Plate objective scoring tool (POST)** (Fig. 56.8): **A** = distal mucocutaneous junction, **B** = glanular knob where the plate changes direction, **C** = glanular–coronal junction. **AB** = neomeatus length, **BC** = glans fusion length. A **low AB/BC ratio means a poor plate**, which favours **augmentation (inlay graft) over tubularised incised plate (TIP) repair**. Decide **before incising**.
- Circumscribe → deglove partially or fully by curvature → create a **mucosal collar**, rotating inner preputial tissue ventrally for skin cover and cosmesis.

Assessment and Management of Ventral Penile Curvature

- Curvature (chordee) occurs with or without hypospadias. **It is the main determinant of one-stage vs multistage repair.** In adults, even **20–30° ventral curvature** can impair intercourse and satisfaction.
- **Causes:** tethered ventral skin, fibrous dysplastic dartos, a short urethra, corporal disproportion. Devine and Horton (1973) staged by dysplasia of Buck's fascia, dartos and spongiosum. A later **four-group** system: **skin tethering • dysplastic Buck's and dartos fascia • corporal disproportion • congenitally short urethra.**

Assessment: parents' history or photographs of erections → bougie à boule or catheter under the ventral tissue (shows urethral and skin quality) → circumscribe and **deglove to the penoscrotal junction**, excising dysplastic dartos → **artificial erection** (tourniquet at the base plus saline, Gittes–McLaughlin; **or compress the corpora at the base;** or prostaglandin). **A tourniquet can hide basal curvature.**

- **Grading:** mild <30°, moderate 30–45°, severe ≥45°.
- A **goniometer** reveals that “eyeballing” **underestimates** curvature, and apps can capture images. But goniometer interrater reliability is **poor** (an acute angle and an arc behave differently), so a reliable standard is still needed.

Algorithm (Fig. 56.9): deglove + artificial erection → straight → urethroplasty. **Residual <30° → dorsal plication.** >30° → **divide the urethral plate** → still >30° → **ventral corporal lengthening**. Poor ventral skin despite <30° → consider lengthening.

Dorsal shortening (plication)	Ventral lengthening (corporotomy ± graft)
Nesbit (1965): excise paired ellipses of dorsal tunica and close transversely. Baskin: a 12 o'clock nerve-free zone allows a single midline plication. Authors: midline dorsal incision to corporal tissue, closed transversely (Heineke–Mikulicz). Some plicate over a longer distance for an arc Simpler; keeps ventral skin options Risks: neurovascular injury off the midline (elevate the bundle); recurrence, especially at puberty; less effective in severe or arc curvature Authors: ≤15°; many: up to 30°	“Fairy cuts” (superficial, not through the tunica) · transverse corporotomies (one to three, full thickness, ungrafted, as in the staged tubularised autograft (STAG) repair) · authors' preference: single full-thickness corporotomy from 9 to 3 o'clock (or until mild reverse hinge) with watertight graft: dermis, tunica vaginalis flap or graft, or non-autologous material (AlloDerm, small intestinal submucosa). Dermal graft 20% larger than the defect Adds length; better for severe curvature. No technique has been compared head to head Risks: longer, more complex; limits later ventral skin options; theoretical erectile dysfunction (not seen in the authors' adults) Moderate or severe curvature after degloving and plate division

- **Evidence:** in severe perineal hypospadias, **recurrent curvature followed plication in about a fifth but no staged grafted repairs**. In a 100-patient series, **recurrence was 28% after plication vs 9% after grafting**. **Not dividing the plate is an independent risk factor**.
- **Curvature may worsen at puberty**, or represent under-correction. Correct it fully the first time, because persistent curvature needs multiple staged revisions.
- **Congenital penile curvature** (severe chordee, **normal urethra**): disrupted tunical collagen and matrix. Algorithm: deglove → plicate → severe cases transect the urethra, graft and stage. **Authors' alternative: mobilise the normal urethra in situ**, corporotomy and graft in **one operation** (Fig. 56.13). **Recurrence 2–15%**, mostly after plication.
- **Dorsal or lateral curvature:** usually **multipoint plication**; rarely a patch graft with urethral or neurovascular mobilisation. Plication fails more often for lateral, dorsal or spiral curves (thinner lateral tunica).

KEY POINTS

- **Curvature <30° → dorsal plication**, without clinically apparent shortening.
- Objective measurement is elusive, but **curvature assessment drives technique** and must be fully corrected.
- **Plate transection and/or ventral grafting** are for moderate or severe curvature after degloving and dartos excision.
- **Under-corrected curvature causes major morbidity** and repeated surgery.

KEY POINTS

- **Goals:** normal voiding, sexual function and appearance.
- Counselling weighs **early anaesthesia** against **early-repair benefits**.
- **No routine imaging** in non-syndromic hypospadias.
- **Preoperative androgens** transiently enlarge the penis; their effect on complications is **controversial**.

- **Examination findings to plan for:** deficient ventral prepuce, downward glans tilt, deviated median raphe, ventral curvature, scrotal encroachment, scrotal cleft, penoscrotal transposition.
- **Regional block is important. Caudal block reduces anaesthetic and does not appear to increase complications.**

Concept — *Curvature, not meatal position, sets the operation. Mild curvature is shortened dorsally in one stage; severe curvature is lengthened ventrally and usually staged. Undercorrection is the error that keeps generating further surgery through puberty.*

General Considerations of Surgical Repair

- **Age:** repair under 6 months had more success in some studies but not others. A systematic review (over 16,000 patients) found **fewer complications in younger children** and recommended repair **by 18 months**. Surgery after 18 months carried no long-term emotional harm. **Authors: start at 6 months**, especially if staged, within the **AAP range of 6–18 months**. Adult repair outcomes were once worse; recent series show no excess.
- **Antibiotics:** early studies counted asymptomatic bacteriuria. One randomised trial in TIP found **fewer complicated UTIs and complications with continuous antibiotic prophylaxis (CAP)**; others found no effect, and some question preoperative antibiotics altogether. **Infection is rare**, so the question needs a multicentre trial. Authors' practice ranges from **none to 3 days starting the day before stent removal**.
- **Sutures:** must hold until healed but **dissolve before suture tracks epithelialise** (fistula, sinus), with little inflammation. **Rapidly absorbed chromic or polyglactin did better than slowly absorbed polydioxanone (PDS)** in urethroplasty. Size: **6-0 to 8-0 in infants; 5-0 in adults**.
- **Learning curve:** **higher surgeon volume and more years after fellowship → fewer complications**.
- **Never use hair-bearing skin** for urethroplasty: postpubertal hair → **stones and recurrent UTI** (Fig. 56.14).
- **Technique:** gentle handling (fine forceps, skin hooks, stay sutures), **magnification**, generous flap mobilisation, and **tension-free urethra, glans and skin**. Tension → reduced capillary flow → ischaemia → fibrosis, contraction, infection and breakdown.
- **Closure technique** is debated: epithelial inversion and two-layer closure have been favoured, but simple interrupted full-thickness closure has also done well.
- **Barrier layers** of vascularised, non-epithelialised tissue over the neourethra **reduce fistula** (dartos interposition after TIP), provided they do not compress the glansplasty. Sources: **de-epithelialised dorsal preputial dartos (Belman flap**, harvested after rotating Byars flaps), de-epithelialised ventral shaft skin, **tunica vaginalis** (mobilised to the external ring to avoid tethering), spongiosum. Secure laterally so suture lines are **offset**.
- **Haemostasis:** pinpoint cautery. Authors use an **intermittent tourniquet, released every 30 minutes**. Haematoma impairs flap and graft healing.
- **Dressing:** a **circumferential Tegaderm from glans to penoscrotal junction**, including the glans (prevents swelling), with **no constricting proximal ring**. Or a **“sandwich” dressing** (stent between two Tegaderm layers) for **72 hours**. Then petroleum jelly; warn parents about oedema and bruising.

KEY POINTS

- **Higher surgical volume → fewer complications.**

- **Non-overlapping suture lines, tension-free closure and vascularised barrier layers** improve success.

Concept — *Hypospadias surgery succeeds or fails on **blood supply**. Every technical rule (fine instruments, no tension, offset suture lines, vascular barrier layers, quickly absorbed sutures, no haematoma) protects perfusion of thin, dysplastic tissue.*

Distal Hypospadias Repair

- **More than 200 techniques** exist, mostly variants. The tenets: **correct curvature** → **advance the urethra** (with or without urethroplasty) → **glansplasty** → **skin closure** ± **preputioplasty**.
- Urethroplasty sources: **local genital skin flaps**; **preputial or extragenital flaps or grafts**; **tubularisation of the plate**.
- **TIP is used for over 90% of distal and about 80% of midshaft repairs**.
- Distal repair suits minimal curvature and favourable anatomy for **single-stage** repair; more severe variants are treated as proximal.

ADVANCEMENT PROCEDURES

- **No tubularisation**; for a **distal, glanular meatus with minimal or no curvature**.

ADVANCEMENT PROCEDURES COMPARED

Procedure	Indication	Technique	Pitfall
Urethromeatoplasty	Stenotic distal meatus with a blind pit and partly fused glans	Heineke–Mikulicz : vertical incision, horizontal closure, flattening the posterior plate	—
Meatal advancement and glanuloplasty incorporated (MAGPI) (Duckett, 1981)	Meatus on the glans only	Advance the meatus distally without tubularising; glanuloplasty in two layers; mucosal collar	Used more proximally → tension → retrusive meatus and flattened glans ; now used selectively
M inverted V (MIV) glansplasty (Decter, 1991)	Proximal glanular or coronal meatus ; compliant parametatal skin; adequate glans with ventromedial upward angulation	M-shaped marks → distal meatal traction → ventral dissection → meatoplasty → two-layer glans closure → mucosal collar	Designed to avoid MAGPI retraction; thin ventral skin, so do not breach the urethra
Urethral advancement (sleeve) (Koff)	Selected distal cases	Mobilise the distal urethra, sometimes aggressively, and advance it tension-free into the glans tip; excise the midline spongiosum to seat it	Devascularisation → stricture ; retraction under tension

Concept — *Advancement works only when **the meatus has a short way to travel**. Each technique failed when stretched to more proximal meati, because tension produced retraction. The history of MAGPI teaches that **indication matters more than technique**.*

TUBULARIZATION TECHNIQUES

- A formal urethroplasty, for a meatus too proximal to advance, when curvature is mild or resolves with degloving. Success depends on **glans, plate and periurethral tissue quality**.
- **Always check the posterior plate for a transverse cleft** between meatus and groove. **It must be flat** or the stream is deflected.

TUBULARISATION AND FLAP TECHNIQUES FOR DISTAL HYPOSPADIAS

Technique	Indication	Key steps	Strengths and concerns
Glans approximation procedure (GAP) (Zaontz, 1989); pyramid = GAP with intact prepuce	Proximal glanular or coronal; wide, deep groove ; non-compliant “fish-mouth” meatus (often MIP)	Deep glans incisions, wing mobilisation, two-layer neourethra, dartos barrier	Pinching the wings together would give a normal look
Thiersch–Duplay (TD) (Thiersch 1869, Duplay 1874)	Distal; adequate glans width	U-incision lateral to the groove sized to 12 Fr → deep, wide wing mobilisation → double running subcuticular neourethra under magnification → de-epithelialised dartos flap (double flap if more proximal) → two-layer glans closure → mucosal collar	Tubularises local tissue distal to the meatus; the authors use it often
TIP (Snodgrass, 1994; Rich's hinge, 1989)	Narrow plate otherwise too tight to tubularise	Deep midline plate incision down to the corpora → tubularise distal to proximal, stopping 2–3 mm short of the planned meatus (oval opening) → dartos cover → glansplasty	Slit-like meatus ; most used worldwide. Concern: incision contracts (narrowing, stenosis). Authors: incise the plate first, measure, then add lateral incisions to 12 mm total width ; incise only for a transverse cleft; often add an inner preputial inlay graft
Mathieu (flip-flap)	Deficient distal plate	Parametatal proximally based flap flipped onto the plate	Narrow flap base → ischaemia, fistula, stenosis ; fish-mouth meatus. Slit-like adjusted Mathieu (SLAM) : wide-based flap preserving spongiosum + V-excision → slit meatus
Onlay island flap (OIF)	Narrow plate	Dorsal inner preputial island flap on its pedicle, transposed ventrally onto the plate (authors: ventral shaft incision to the penoscrotal junction exposes the pedicle; harvest from the penopubic junction distally)	Popular in the 1990s, then displaced by TIP. Fish-mouth meatus without Rich's modification; diverticulum (elastic preputial skin: do not make the flap too wide); less stenosis

KEY POINT

- Many distal repairs are variants of one another. **Use the technique you are comfortable with and that gives a high success rate.**

Figure legend, Fig. 56.21 (after Snodgrass, 2005) — In TIP, the **first stitch is about 3 mm proximal to the end of the plate**, creating an oval meatus.

Concept — Distal tubularisation is a trade between **tissue and tension**. Incising (TIP) widens a plate in place at the risk of scar contraction. Adding tissue (Mathieu, onlay, inlay) avoids tension at the cost of meatal shape and diverticulum. The choice follows plate width and depth.

Proximal Hypospadias

- Recurrent curvature after dorsal plication in severe cases has shifted practice towards **ventral corporal lengthening and therefore multistage repair**.

One-Stage Hypospadias Repair

- Used less often: limited to curvature resolving with degloving or plication, with favourable tissue. Tissue options: **preputial skin, local skin, buccal graft**.

ONE-STAGE PROXIMAL REPAIRS

Technique	Indication	Points
Transverse preputial island flap (TPIF, "Duckett tube") (1980)	Severe curvature corrected by skin release and plate division , or after corporal grafting	Authors anchor one flap edge to the ventral shaft, then close the tube with Lembert sutures to an even calibre (12 mm), with the pedicle over the suture line. The anchoring point removes laxity (less diverticulum). Tissue in the glans gives an oval meatus . Higher complications led Duckett himself to prefer OIF
OIF (proximal use)	Curvature corrected by degloving; plate kept but too thin to tubularise	Dissect dorsally above Buck's fascia, preserving the pedicle. Combined plate plus flap width ≤ 12 mm ; narrow the proximal flap to avoid diverticulum at the native-neo junction
Proximal TIP	Curvature corrected with skin dissection or plication; adequate plate	Plate mobilisation and transverse corporotomies were added for $>30^\circ$ curvature. But aggressive plate mobilisation devascularises $\rightarrow$ stricture , and there is more recurrent curvature , so it is abandoned in favour of staged Byars or STAG

- **Why staging:** correct curvature at stage one, urethroplasty at stage two, giving supple ventral tissue. A **urethroplasty over a fresh corporotomy and graft** fails more often.
- Advocates of single-stage repair argue that fewer children overall undergo multiple operations. Single-stage **tunica vaginalis lengthening plus TPIF** has been reported.

Two-Stage Hypospadias Repair

- Now the **approach of choice for most proximal hypospadias**, in three categories defined by the urethroplasty.

TWO-STAGE REPAIR CATEGORIES

Category	Stage 1	Stage 2 (≥ 6 months)	Notes
Free graft: Bracka (inner prepuce or buccal); STAG	Correct curvature (STAG: one to three ungrafted ventral corporotomies); divide the plate ; midline glans incision as the bed; quilted graft + compression dressing	U-incision (TD), tubularise, glansplasty, multilayer (tunica vaginalis) cover	Graft over corporotomies can contract $\rightarrow$ STAC : no skin transfer at stage 1, graft at stage 2, tubularise at stage 3. STAG adds proximal urethral mobilisation and wider glans dissection
Pedicle flap: Byars (1951)	Dorsal preputial skin split and transposed ventrally on its pedicle (midline-joined, or as a single island flap); correct penoscrotal transposition (may need three stages)	Long U-incision, TD tubularisation, watertight two-layer closure, even calibre	Build a supple dartos bed over the corporoplasty at stage 1
Distal-first (Ulaanbaatar; TIP-type distal closure)	Correct curvature and close the distal urethra and glans with an interposition flap or graft; the child voids from a proximal meatus	Join the proximal urethra to the closed distal segment	Ulaanbaatar : transverse island tube tunnelled into the glans

Figure legend, Fig. 56.30 (Jayanthi, 2017): modified Ulaanbaatar staged repair — First stage: correct curvature with **small intestinal submucosa** → **transverse island Duckett tube** for the glanular and distal penile urethra, tunnelled or glansplastied → **Byars flaps** bridge to the native urethra → drip stent. Second stage **at 6 months**: elliptical incision around the defect and tubularise. (The example case had 46,XY ovotesticular DSD.)

- **At stage 2**, check graft or flap take. With skin deficiency or tethering, place an **intermediate dorsal inlay buccal graft** and close later, or do a dorsal buccal inlay with same-stage urethroplasty (Fig. 56.31).
- **Always repeat artificial erection before urethroplasty**: plicate minor recurrence, or lengthen again.
- **Interval ≥6 months** (authors **8–12**), checking take, recurrence and skin suppleness. **Graft massage** (petroleum jelly, vitamin E, silicone gel) from **2–4 weeks** is unvalidated but low-risk.

Figure legend, Fig. 56.31: single-stage buccal inlay — When the plate is **too narrow to tubularise**: U-shaped TD incision → **midline plate incision** → **buccal graft quilted into the midline** → urethroplasty to the U width → barrier layers → closure.

Comments on Proximal Hypospadias Repair

- Historical series **underestimated** proximal complications by pooling mild and severe variants.
- **25–75% need additional procedures after a planned two-stage repair**, each adding further risk.
- **Distal hypospadias is corrected successfully in about 80%**; the specialty's effort must now go into **reducing complications in proximal hypospadias**, where each additional procedure adds further risk.
- **Longer urethroplasty = more demanding**: a long neourethra of uniform calibre is hard to build, and the neourethra may be **inherently narrowed and unable to expand** to propel the urine bolus.
- **Poiseuille's law**: resistance to flow is **proportional to tube length** and **inversely proportional to the radius to the fourth power**. Small changes in calibre therefore have a large effect, and **the longer the tube, the greater the stricture risk**.
- Failure of the neourethra to expand → **raised resistance** → **fistula and/or diverticulum**.
- **The glans is non-distensible** (dense spongiosum). Even with uniform calibre, the **fixed distal urethra** causes turbulence and **high proximal pressure** → **diverticulum**.
- The neourethra is a **skin-lined tube without spongiosum**, an independent risk for progressive distension.
- Tissue **dysplasia**, the **severity of curvature** and **skin shortage** may impair healing, perfusion and growth with age. Each surgeon's numbers are falling, so **multi-institutional collaboration** is essential.

	TIP, long penoscrotal neourethra	OIF, penoscrotal neourethra
Tissue	More rigid neourethra	More pliable, elastic preputial tissue
Resistance	Fixed resistance along a long length	Lower: distends with micturition
Uroflow	Prolonged, flattened void; risk of proximal fistula	More normal flow pattern

Figure legend, Fig. 56.32 (Braga et al., 2007) — A short coronal TIP neourethra gives a near-normal flow curve, because there is little resistance over a short distance. The meatus is assumed patent in all cases.

KEY POINTS

- **Proximal hypospadias has a higher complication rate** than distal.
- **The trend is towards staged repair** for severe hypospadias: **curvature is corrected at stage one, urethroplasty at stage two**. This gives supple ventral tissue for tubularisation and avoids a urethroplasty laid over a corporal graft in one stage.

- **25–75% need additional procedures** after a planned two-stage repair.

Concept — Proximal repairs fail for **physical reasons**: a long, rigid, spongiosum-free tube ending in a fixed glans. Length and stiffness raise resistance, and raised resistance produces fistulae, diverticula and strictures.

Postoperative Management

- A careless postoperative course can ruin a good repair, yet **quality evidence for specific regimens is lacking**.
- **First 48 hours**: urinary diversion plus a **compressive dressing**, which reduces stress on the neourethra, tamponades bleeding and limits swelling. The authors leave the dressing **72 hours**, then apply petroleum jelly.
- **A soft, pliable urethral stent in complicated repairs reduces meatal stenosis and fistula**. A **suprapubic tube** is used in older children and redo cases.
- **Diversion lasts 3–21 days**, by complexity, urethroplasty length and preference. Longer stenting reduced complications in one study but is not universally accepted, and must be weighed against **discomfort and UTI risk, which rises with each day**. Short distal repairs have been done **without a stent** with no change in outcome.
- **Bladder spasm** → high-pressure voiding around the stent → stress on the repair, so give **oxybutynin until the stent is removed**.
- **Long, stiff stents** press on the ventral repair and risk **glans dehiscence**. A **Foley catheter must be fixed in the midline to the anterior abdominal wall**. Tape often slips and pulls the catheter laterally or ventrally, so the authors have **abandoned Foley catheters** because of glans dehiscence.
- **Analgesia**: paracetamol is near-universal, and about half of urologists add **non-steroidal anti-inflammatory drugs (NSAIDs)**. **Ibuprofen and ketorolac do not increase bleeding**. **Regular paracetamol plus ibuprofen** minimises opioid use, tapering after **48–72 hours**.

Complications of Hypospadias Repair

- Duckett described “hypospadiology” as a **humbling blend of art and science**. The tissues are delicate and healing depends on **fragile neovascularisation**, threatened by oedema, infection and variable healing.
- **Complications have lacked standard definitions**, which makes reporting inconsistent. The standardised set is in Table 56.1.

TABLE 56.1 COMPLICATIONS AND DEFINITIONS

Complication	Standardised definition
Urethroplasty complications (primary outcome)	
Fistula	Urethral leak anywhere below the meatus
Glans dehiscence	Complete separation of the glans wings → coronal or more proximal meatus , or wings separated by a skin bridge; objectively glans fusion <2 mm
Meatal stenosis	Obstructive symptoms (straining, prolonged voiding, UTI, retention) plus calibration <8 Fr before puberty or <12 Fr after puberty
Urethral stricture	Obstructive symptoms plus near-closure of the urethra on urethroscopy
Urethral diverticulum	Segmental sacculation during voiding
Recurrent curvature	Residual ventral, dorsal or lateral curvature >30° on erection (spontaneous or artificial)
Skin complications (secondary outcome)	
Skin surplus	Excess skin ≥2 cm in a circumcised patient with the suprapubic fat pad retracted

Complication	Standardised definition
Penile torsion	Glanular plane >30° from the vertical ; specify direction
Non-retractable reconstructed prepuce	Meatus not visible (without lichen sclerosus); acceptable before puberty, abnormal after
Preputial dehiscence	Defect recreating an incomplete prepuce
Preputial fistula	Fistula-like preputial defect not communicating with the urethra
Lichen sclerosus	White cicatrix, preferably confirmed histologically
Also listed	Skin deficiency with tethering; postpubertal phimosis; inclusion cyst

Courtesy Warren Snodgrass, MD.

- **What the table is for:** a common language, so that series can be compared. The pattern is that **urethroplasty failures are the primary outcome, skin problems secondary**. The numbers to hold are **>30° for curvature and torsion, <8 Fr / <12 Fr for stenosis and <2 mm glans fusion**. The trap: **calibration alone is not stenosis**; obstructive symptoms are also required.

Urethrocutaneous Fistula

- **The commonest complication: just under 10%** at short-term follow-up. It occurs **most often at the original meatus or the coronal margin**, from pinpoint to large.
- **Causes:** ischaemia, oedema, infection, haematoma and especially **distal obstruction** (meatal stenosis or stricture → high pressure and turbulence). **Technical causes:** overlapping suture lines, inadequately inverted epithelium, poorly absorbable sutures. Risk **rises with severity**.
- It is noticed as **multiple streams**, mostly **within 2 years**, or at toilet training.
- **Small early fistulae rarely close spontaneously. Repair at least 6–12 months** after the primary operation, once tissues soften.

Assessment under anaesthesia: bougie à boule calibration for patency and irregular calibre → **urethroscopy** if irregular → **“stress test”**: catheter in the urethra, compress proximally and distally, inject saline or methylene blue to reveal hidden tracts.

FISTULA REPAIR BY SITE AND SIZE

Fistula	Repair
Small, shaft, proximal to the corona	Excise the epithelialised tract → subepithelial urothelial closure → multiple offset layers
Large, or thinned tissue	Trap-door or island flap of shaft skin; second layer or vascularised flap, offset
Coronal margin	Implies distal narrowing or poor tissue → revise the urethroplasty , not simple excision

- **Recurrence about 20%** even after good repair. Reduce it with **interposed vascularised non-epithelialised layers**: redundant dartos, **scrotal tunica vaginalis flap**, scrotal dartos, a de-epithelialised flap, or the **Cecil modification** when tissue is scarce.
- **Diversion:** none for simple repairs; **7–14 days** for complex redo repairs.

Figure legend, Fig. 56.34 — In fistula repair after a two-stage proximal repair, the **distance from the glans drives decision-making**. The sequence is: mark around the tract and raise flaps for barrier cover → isolate the tract → undermine → primary closure → dartos over the repair → skin closure. There was no recurrence at 6 months.

Glans Dehiscence

- **Causes:** catheter tension, poor technique, **glans closure under tension**, vascular compromise. The meatus **retreats to a coronal or subcoronal position**, in effect a breakdown of the repair.
- **Reoperate only for voiding symptoms.** If repair is needed, an **interposition inlay graft**, as a staged step or in a single stage, avoids glans tension.

Meatal Stenosis

- Definitions vary. **Risk:** urethral closure carried **too far distally**, or a **tight glans closure**. Longer stenting may help (weak evidence).
- **The postoperative meatus should calibrate ≥ 8 Fr, increasing with age.** Clinically relevant stenosis means **stranguria, deflected stream, poor flow, retention** and a **“laser-beam” stream**.
- **Escalation:**
 - **Sound dilatation $\pm$ topical betamethasone** in some cases.
 - **Routine postoperative dilatation is unnecessary**, but early dilatation may reveal subclinical stenosis.
 - **Refractory to dilatation $\rightarrow$ dorsal midline meatotomy** (rarely), otherwise **redo urethroplasty**.
- **Meatal retrusion:** excessive distal tension, or after glans dehiscence. It results from poor planning: an immobile or poorly vascularised urethra retracts. **Revise if the deflected stream prevents standing voiding.**

Urethral Stricture

- Rates depend on **technique and urethroplasty length**. Sites: **native-neourethral junction**, along the neourethra (especially at the **coronal margin with a narrow plate**), or within the glans. It presents with a weak stream, stranguria, retention or UTI.
- **Causes:** technique, ischaemia, trauma, infection. **Characterise by cystoscopy** (length, calibre, site).
- **Short, mildly symptomatic strictures $\rightarrow$ dilatation or endoscopic incision, rarely durable. Internal urethrotomy** succeeds poorly and can worsen scarring. Even **strictures < 1 cm with intermittent catheterisation** seldom keep normal flow. Results may be better for short anastomotic strictures treated at 3 months, but the authors rarely see success.
- **Refractory or extensive strictures $\rightarrow$ revision urethroplasty.** Expect tissue loss, so be ready to harvest **local flap or buccal mucosa**; this is often multistage.
- **Hair-bearing skin in the urethra** after multistage or complex repairs, combined with narrowing, becomes a **nidus for stones and UTI**. Options: laser epilation; endoscopic stone removal. When **stones, hair and UTI coexist $\rightarrow$ resect and replace** with buccal graft or local skin.
- **Recurrent UTI $\rightarrow$ look for diverticulum or stricture.** An **enlarged prostatic utricle** causing stasis $\rightarrow$ **excise or fulgurate**.

Urethral Diverticulum

- It presents with a weak stream, **post-void dribbling**, UTI, rarely haematuria, **ballooning of the shaft**, or the need to **“milk” the urethra**.
- **Commoner after preputial flap, two-stage and proximal repairs:** there is **no spongiosum**, and a **rigid glanular neourethra** raises penile urethral pressure.
- **Small saccular diverticulum $\rightarrow$ excise and reduce to a uniform calibre. Extensive diverticulum (more common) $\rightarrow$ excise redundant tissue, close the urethra and add multilayer reinforcement.**
- The redundant wall is **well vascularised** and can be **reused as a flap** to repair coexisting fistula or distal stricture. **Extraurethral plication** avoids entering the urethra.

Figure legend, Fig. 56.36 — Excise the excess epithelium but **keep the redundant dartos** as an extra layer over the urethral closure.

Balanitis Xerotica Obliterans

- **Balanitis xerotica obliterans (BXO), or lichen sclerosus, is a chronic inflammatory dermatosis.** It occurs with phimosis in uncircumcised boys, with meatal stenosis, or with progressive urethral narrowing after repair, appearing **pale and white**.
- **Meatal disease only** → **topical betamethasone or clobetasol**, or **systemic tacrolimus**. These **usually fail**.
- **Definitive: resect all involved tissue**, send it for histology, and replace with **inner prepuce or buccal mucosa in a staged repair**. **BXO can invade buccal mucosa**.
- **Recurrence up to 40%** on medium-term follow-up, so **monitor closely**.

BXO sequence: urethral “**cutback**” to healthy tissue → topical steroids → excise persistent disease → place a **buccal graft** and confirm take without recurrence → urethral closure.

Recurrent Penile Curvature

- It damages urinary and sexual function and may **worsen at puberty** through differential dorsal–ventral corporal growth.
- **Causes: curvature underestimated or incompletely corrected, graft contraction** after corporotomy, **ventral skin traction**, or a preserved **tethering plate**.
- **Assess by complete degloving with artificial erection**. A **tourniquet can hide basal curvature**, so compress the corpora against the pubis instead.

Staged correction (counsel the family that several procedures are likely): stage 1 identifies the cause (watch for **corporal disproportion after a previous dorsal plication**). If a corporal graft is used, cover it with **healthy dartos as a bed for buccal graft** → stage 2: **ventral buccal graft** unless skin allows tension-free cover → stage 3: **Cecil modification** if shaft skin is deficient → stage 4: tension-free skin closure.

Figure legend, Fig. 56.38 — A 14-year-old after **infant single-stage repair with dorsal plication** presented with marked ventral curvature. Correcting it left a **long urethral gap**, which needed a buccal inlay and then final closure.

Skin Complications

- **Buried penis with shortening** from concealment and poor skin fixation is prevented by **degloving plus anchoring sutures from Buck's fascia to the dermis** at the penopubic and penoscrotal junction.
- **Suture sinus tracts** come from thin ventral skin, full-thickness sutures, or sutures tied too tight (ischaemia).
- **Penoscrotal transposition or webbing** after proximal repair → **scrotoplasty** to prevent concealment.

Concept — *Almost every urethroplasty complication traces back to **distal obstruction, ischaemia or tension**. That is why fistula and diverticulum repair start by excluding distal narrowing, and why every repair adds vascularised barrier layers.*

HYPOSPADIAS OUTCOMES

- The literature is **weak**: retrospective, small, technically varied, with poorly defined outcomes, no independent assessment and much loss to follow-up.
- **Largest meta-analysis of TIP repair** (Pfistermuller et al., 2015; 49 studies, 4,675 boys, mean follow-up 16 months): **complications 10.6%, reoperation 4.5%; fistula 5.7% > meatal stenosis 3.6% > stricture 1.3%**. An **extra dartos layer improved outcomes; proximal and redo cases did significantly worse**.

- **TIP vs Mathieu (meta-analysis): similar overall complications and fistula rates, but meatal stenosis much commoner after TIP.**
- **Recurrent UTI after repair is uncommon (about 2%).** Imaging finds distal narrowing, VUR, diverticulum or an enlarged utricle. **TIP → raised post-void residual; OIF → diverticulum or fistula. Image anatomically any child with recurrent UTI.**
- **Distal hypospadias: success 83–95%. Complications in large single-centre series are higher with longer follow-up and a more proximal meatus. After TD repair, fistula and glans dehiscence occur early (first year) but stricture late.**
- **Proximal hypospadias: complications 23–68%; fistula is again commonest (3–45%), then glans dehiscence and meatal stenosis. Diverticula are commoner after Byars flap or onlay than after two-stage graft repair (STAG or Bracka).**
- **Recurrent curvature after two-stage repair is low (0–10%).** Two-stage repair with corporoplasty gives **greater penile length and better cosmesis.** Penile length is the chief concern of adults repaired in infancy.
- **STAG: complications <25%, mainly glans dehiscence or fistula; recurrent curvature low. Extra layers and better blood supply reduce complications.**
- In a four-way comparison (TIP, preputial island onlay, island flap tube, two-stage Bracka), **the approach did not change complications. Dorsal plication plus proximal urethral mobilisation** corrects most curvature and avoids laying a skin graft over a corporal graft.
- **Byars flap two-stage repair: high complications (30–70%), mainly fistula then glans dehiscence.** Diverticulum results from **no spongiosum plus no fixation of the flap** to the ventral shaft.
- In one centre, **single-stage repairs had more complications than planned two-stage repairs** (not significant). Two-stage approaches that close the distal urethra and correct curvature at stage one report favourable results.
- Penile perception scores were **better with staged repair**, which gave a **longer penis**, with similar complications and urinary complaints across four techniques.
- **No direct comparison** identifies the ideal approach to proximal hypospadias with severe curvature, and **long-term data are lacking.**

FIG. 56.39 SUMMARY OF RECENT STUDIES ON PROXIMAL HYPOSPADIAS OUTCOMES — REBUILT BY TECHNIQUE

Repair	Reported complication rate (follow-up about 1.5–5 years)	Note
Two-stage Byars flap	43–81% across four centres	Highest-risk staged option
Long single-stage TIP	About 60%	—
Single-stage (mixed)	About 60%	—
Dorsal inlay graft	About 30–50%	—
Pedicled preputial flap	About one-third	—
Two-stage repair (graft)	About 38%	—
Proximal TIP	About 25%	No longer recommended: higher recurrent curvature
STAG	About 23%	Lowest reported
STAC	Unpublished	The authors' current treatment of choice

Data from Stanasel et al., 2015; McNamara et al., 2015; Pippi Salle et al., 2016; Long et al., 2017; Snodgrass et al., 2017, 2021; Chan et al., 2021.

PROXIMAL HYPOSPADIAS COMPLICATION RATES: FIGURE VERSUS BODY TEXT

Series	Fig. 56.39	Body text	Reason
Boston, two-stage Byars	53% (follow-up 40 months)	49% (follow-up 46 months)	Different data cut or denominator
Toronto, long TIP / staged	61% / 38%	53% / 32% (follow-up 30–48 months)	Different denominator (140 patients in the text vs per-technique counts in the figure)

- **What the figure tests: Byars-flap two-stage repairs carry the highest complication rates, and graft-based staged repairs (STAG) the lowest.** Every proximal technique still fails in roughly a quarter to two-thirds. The trap: “complication” is often defined as **any unplanned extra procedure**, so rates are not directly comparable.

Outcome Assessment

- Fistulae are obvious; **early stricture is subtle**. Watch the stream in clinic or on **family video**. Office calibration is difficult, so **examine under anaesthesia** if concerned.
- **Uroflow** helps, but normal values after repair are poorly defined, it cannot be done before toilet training, and **flow improves with age**.
- **Only about half of complications appear in the first year**; about three-quarters are found by 3 years. **Fistulae appear early and strictures late**.
- **Authors' standard:** follow **through puberty**, especially proximal cases. Review after toilet training (stream), and at puberty (velocity, curvature, sexual function).

Patient-Reported Outcomes

- **Surgeon-judged success may differ from patient and parent perception.** Parental **decisional regret** follows complications, hesitancy about surgery and a wish for circumcision. **Better education reduces conflict**.
- **Decision aids** (an online “hypospadias hub” of video and text): families engaged repeatedly, felt confident, and had **less decisional regret**.

HYPOSPADIAS OUTCOME INSTRUMENTS

Instrument	Rated by	Distinguishing feature
HOSE (Hypospadias Objective Scoring Evaluation)	Surgeon	Objective surgical outcome score
Pediatric PPS (Penile Perception Score)	Patient and parents	Satisfaction with length, meatus, glans, skin, straightness
GPS (Genital Perception Score)	Patient	Genital self-perception
HOPE (Hypospadias Objective Penile Evaluation)	Independent panels (parents, patients, urologists) scoring standard photographs against reference pictures	High interobserver reliability

- All share components: meatus, fistula, stream, length, skin, overall appearance. A review found they **neglect sexual function and psychosocial factors** and are **biased towards surgical features** (meatal position and appearance), which may matter less to patients. Uptake of PROs is poor.

TABLE 56.2 HYPOSPADIAS OBJECTIVE PENILE EVALUATION (HOPE) SCORE

Item	10 points	Intermediate	Lowest
1. Meatal position	Position 1	Positions 2–4: 8, 5, 3	Position 5: 1
2. Meatal shape	Normal	Slightly 7; moderately 4	Severely abnormal 1
3. Glans shape	Normal	Slightly 7; moderately 4	Severely abnormal 1
4. Penile skin shape	Normal	Slightly 7; moderately 4	Severely abnormal 1
5.1 Torsion	0–30°	30–50° 7; 50–70° 4	>70° 1
5.2 Curvature on erection	0–30°	30–50° 7; 50–70° 4	>70° 1 (omitted if no erection seen)

From Abbas TO, Front Pediatr 2022. HOPE score = mean points across items 1–5.

- **What it tests:** HOPE scores **appearance on a 1–10 scale per item** (higher = more normal), judged against reference photographs (Fig. 56.41). The pattern: torsion and curvature share the same **30° steps**, and **0–30° counts as normal**, matching Table 56.1's >30° complication threshold. The trap: HOPE is **cosmetic only**, with no voiding or sexual domain.
- **Hypospadias-specific health-related quality of life (HRQoL)** framework (scoping review) defines **five domains: penile appearance, sexual health, psychological and behavioural function, social function, voiding function**. These are the basis for a future patient-reported outcome measure (PROM), because current tools are insufficient (Fig. 56.42).
- **Mild, undiagnosed distal hypospadias** (no curvature) in adults: **minimal sexual, fertility or voiding problems**. Many men are unaware of it, adapt to spraying, are sexually active and father children. **More proximal defects with curvature report worse sexual and functional outcomes**.
- **After surgery:**
 - Older studies reported more behavioural and school problems (worse with severity) and later, less frequent sexual activity.
 - A contemporary controlled series found **equal fertility, sexual satisfaction and age at first intercourse overall**, but **worse satisfaction, fertility and sensation in proximal cases**.
 - Adolescents after proximal repair worry about **penile growth more than meatal position** and are more hesitant about sexual contact.
- **Overall:** no major psychopathology, but **less satisfaction with appearance**, sexual inhibition, more negative genital appraisal, anxiety and lower self-esteem. **Earlier surgery may help**. Lay observers see **distal repairs as normal** and proximal repairs as only slightly different.

KEY POINTS

- **About 60% of urethroplasty complications occur within 1 year**; stricture and curvature can appear late, so **follow proximal cases into puberty**.
- **Risk factors:** proximal meatus, glans width <14 mm, no barrier flap, previous hypospadias surgery.
- **Fistula is commonest:** primary closure if tissue is supple; **redo urethroplasty and glans closure if glans tissue is thin**.
- **Meatotomy or skin flap for stenosis usually fails with BXO**; excise all BXO and stage with buccal graft. Failure is up to 40%, so monitor.

- Long term: more **ejaculatory problems**, less sexual satisfaction, more dissatisfaction with appearance. **PROs are under-used.**

Concept — Hypospadias outcomes are **time-dependent and observer-dependent**. Complications keep appearing for years, and surgeons, patients and parents judge success differently. Only follow-up through puberty with patient-reported measures captures the true result.

Hypospadias Reoperations

- Simple problems (fistula, skin redundancy) can be fixed in one stage. **Each unplanned operation degrades tissue**, scarring the penis, disrupting the blood supply and shrinking options, so **complication rates rise with each redo**.
- **Devine (1973)**: after failure, surgeons tend to attempt “**small operations**”, which make matters worse. What is needed is “**extensive resection of scarring, major shifts of tissues and meticulous reconstruction.**”
- **Assessment**: history of voiding, skin and erectile problems; **previous operative notes** (tunica vaginalis flaps, grafts, urethroplasty type, curvature correction). Have a **low threshold for cystoscopy**, and **always perform artificial erection**, because missed curvature has severe long-term consequences.
- Blood supply of previously operated flaps is unpredictable, so **add layers (tunica vaginalis)**.
- **Tubularisation redo** (Mathieu, TD, redo TIP) has reasonable success. An **inlay graft after incising the plate** improves redo outcomes, but take care with tissue that has already failed.
- **Focal problem (fistula, diverticulum) without curvature** → local or remote skin flaps, as onlay or multilayer cover, especially from **redundant non-hair-bearing shaft skin**.
- **Split-onlay skin (SOS) flap**: a pedicled skin flap of about 1 cm, from anywhere redundant (shaft or circumcision line), **split into an onlay for the urethra plus a second piece for barrier or skin cover**, or de-epithelialised as an extra layer.

ADJUNCTS TO IMPROVE TISSUE IN REOPERATIVE REPAIR

Adjunct	Mechanism	Use and evidence
Topical nitroglycerin	Arterial and (mainly) venous dilatation → more flap blood flow (shown by fluorescein imaging)	Applied for 24 hours after repair; no randomised trial
Hyperbaric oxygen (HBOT)	Oxygen gradient drives angiogenesis ; lowers bacterial load	Established for gas gangrene, radiation necrosis, diabetic foot. Hypospadias data anecdotal ; cost and number of sessions weigh against it
Subcutaneous tissue expansion	Expander under shaft skin, filled via a subinguinal port every 1–2 weeks	Creates non-hair-bearing, androgen-receptor-bearing skin. Largest series: enough skin in 95% , but complications 41% (28% from the expander)

GRAFT MATERIALS FOR SALVAGE URETHROPLASTY

Graft	Advantages	Problems	Status
Extragenital full-thickness skin	Available	Shrinkage, scarring, stricture in single-stage use	Limited
Bladder mucosa	Replaces a scarce plate	Contracts in air → meatal narrowing; prolapse (“cauliflower” meatus) up to 38% ; meatal and junctional stenosis	Largely replaced by buccal mucosa

Graft	Advantages	Problems	Status
Buccal mucosa (cheek)	Adapted to wet and dry, neovascularises, hairless, minimal contraction , low donor morbidity, keeps its histology ; does not contract with growth on a good bed	Donor: mouth tightness, numbness, motor deficit, scar, salivary duct obstruction; more glans dehiscence than lip	Graft of choice
Lip mucosa	Thinner → better in the glans ; fewer complications	Donor morbidity equal to cheek	Preferred within the glans

- **Staged buccal repair beats single-stage** tubularisation (better graft take). A **dorsal inlay** is better than an onlay or complete buccal tube in redo cases with poor blood supply (Fig. 56.31).

Buccal graft technique: harvest **10% larger than the bed** → leave the donor site to heal by secondary intention (closure adds pain, little benefit) → combine sources for long defects → **fenestrate with an 18- or 21-gauge needle** (less haematoma) and keep in saline → incise the recipient bed ventrally from the urethra through the glans to the planned meatus, **preserving dartos** (the blood supply) → suture along its axis without narrowing it → **quilting sutures** against haematoma → wait **≥6 months (up to 12)** before urethroplasty, checking take and contraction → **topical steroid** for contraction (from 1 month) → family “**graft massage**” **from about 2 weeks** (vitamin E oil, silicone gel or petroleum jelly).

- **Results after completion:** **30–75% success**; **meatal stenosis is the commonest failure**, then stricture and breakdown.
- **Cecil-Culp** (Cecil 1946; Culp 1951): urethroplasty → **midline scrotal incision** → recess the repaired penis into the scrotal dartos bed, where it **parasitises the scrotal blood supply** → release at a second stage (Cecil: **6–8 weeks**).
- It fell from favour with single-stage repair and has returned for salvage. The concerns are **hair-bearing scrotal skin** and multiple procedures.
- **Modification:** delay release to **9–12 months**. This transfers more vascularised layers, usually avoids using scrotal skin, and **stretches deficient shaft skin** for tension-free closure.

Figure legend, Fig. 56.44 (Weiss et al., 2018) — After failed two-stage and buccal repairs with two coronal fistulae: the fistulae and glans were closed, and the **ventral skin defect was covered by attaching the shaft to a midline scrotal incision**. About **1 year** later the scar had softened and the skin stretched; after release, the skin was closed in the midline with no recurrence at 3 months.

KEY POINTS

- **Lip mucosa within the glans:** thinner, fewer complications.
- **Buccal mucosa is the graft of choice** in complex reconstruction (properties as tabulated).
- **Cecil modification** vascularises skin after multiple repairs.

Concept — Reoperation is a problem of **blood supply, not technique**. Each step (staging, vascular layers, buccal grafts on dartos, scrotal parasitisation, tissue expansion, nitroglycerin, hyperbaric oxygen) brings vascularised, non-hair-bearing tissue to a bed that previous surgery has impoverished.

FUTURE

- Progress depends on **honest reporting, standardised pre-, intra- and postoperative assessment and definitions**, and **multi-institutional collaboration** to power studies of complication sources.

- **Coaching and teamwork**, as in athletics, can refine technique for rare and difficult cases, particularly proximal hypospadias.
- **Priorities**: lower complication rates in severe hypospadias, **follow-up into adulthood**, and long-term data to set expectations.
- **Open questions**: the ideal meatal position; what curvature is acceptable; how much surgery patients will accept. One patient left with a **coronal meatus but a straight penis and normal voiding is satisfied** (Fig. 56.45). Others are long-term psychological effects and how best to educate families.
- **Technology**: peri-operative **videography and crowd-sourced assessment**; **AI and machine learning** on recorded procedures; genomic, proteomic and microbiome risk profiling; a **central registry** linking anatomy, technique, outcomes and PROs; tissue banking; and perhaps **printed replacement tissue**.